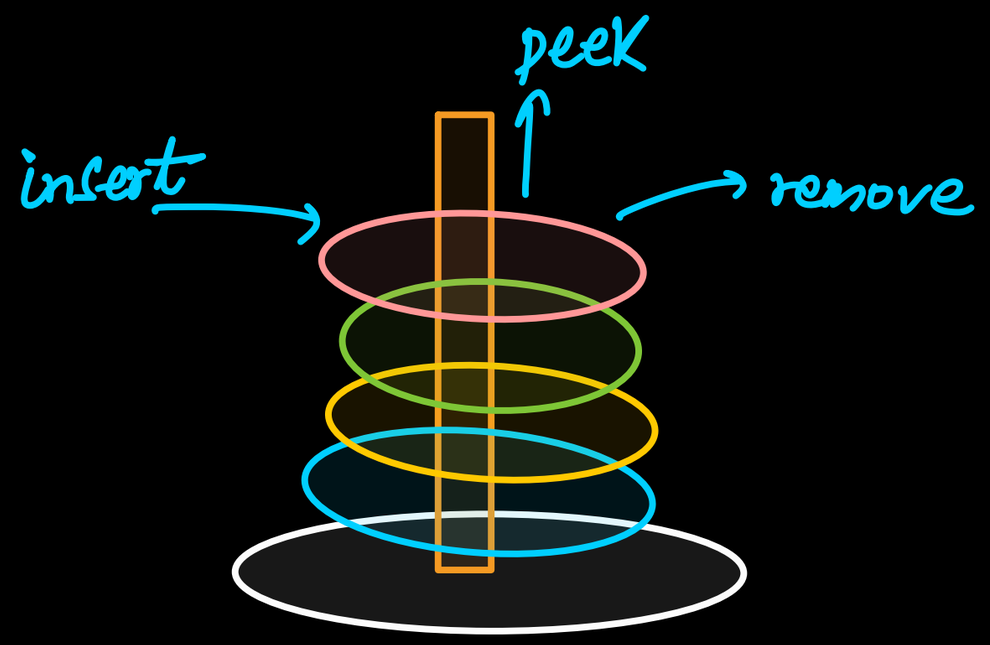
Stacks

First in last out FILO

Last in first out LIFO

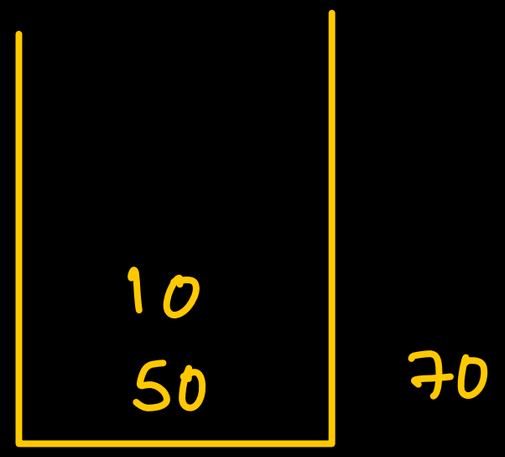
1) Unlimited size

2) Discipline



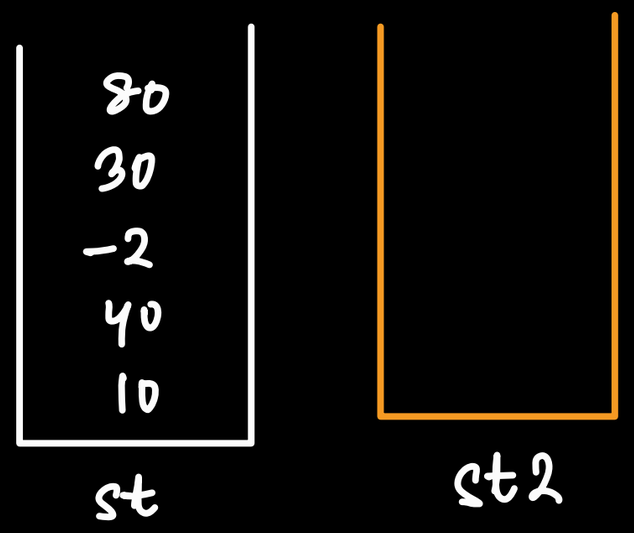
Stack operations

```
Stack<Integer> st = new Stack<>();

st.push(50); //add
st.push(10);
st.push(70);
Sout(st.peek());
st.pop(); //remove
```



push	O(1)
pop	O(1)
peek	O(1)

Traversing the Stack



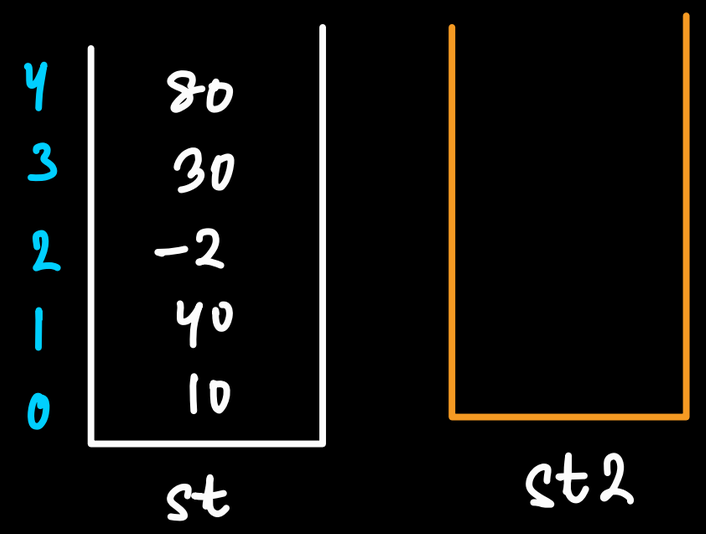

```
while (st.size > 0){
    int top = st.pop();
    sout(top);
    st2.push(top)
}

while(st2.size > 0){
    st.push(st2.pop());
}
```

T.C. = O(n)

A.S. = O(n)

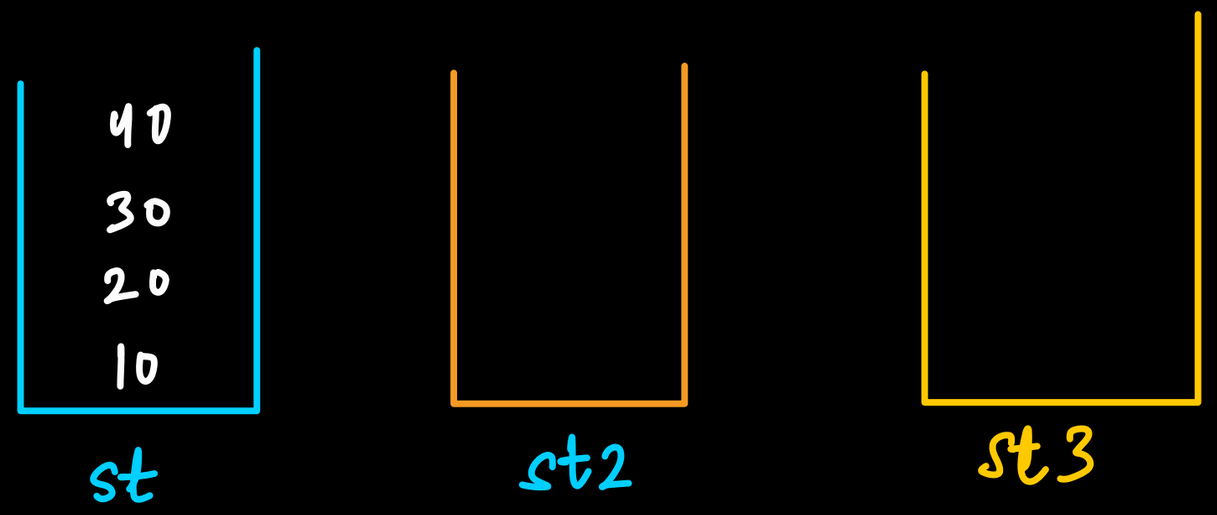
peek, add & remove at any position

get(2)



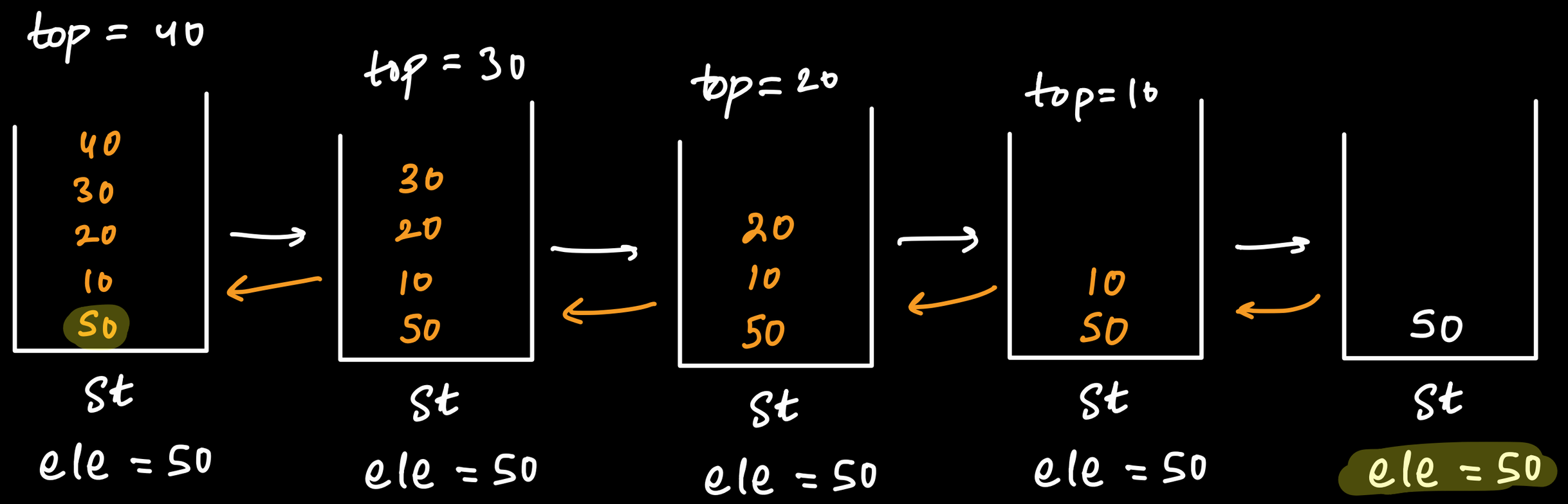

```
while (st.size > idx+1){
    st2.push(st.pop());
}

sout(st.peek());

while(st2.size() > 0){
    st.push(st.pop());
}
```

Reverse the stack

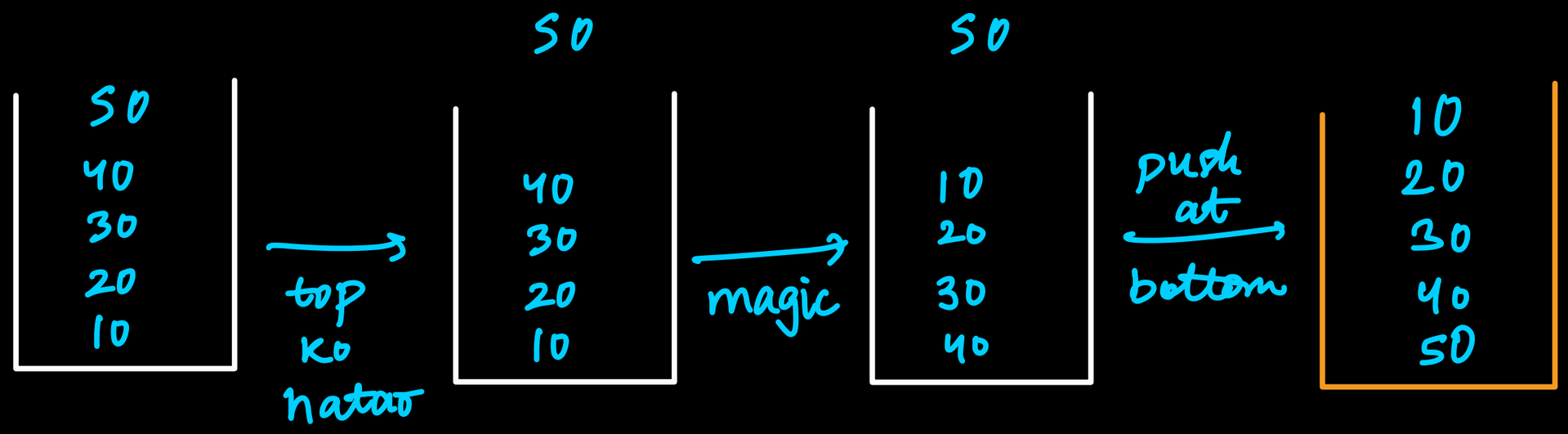


arr = { }

Q. Push element at bottom (Recursively)



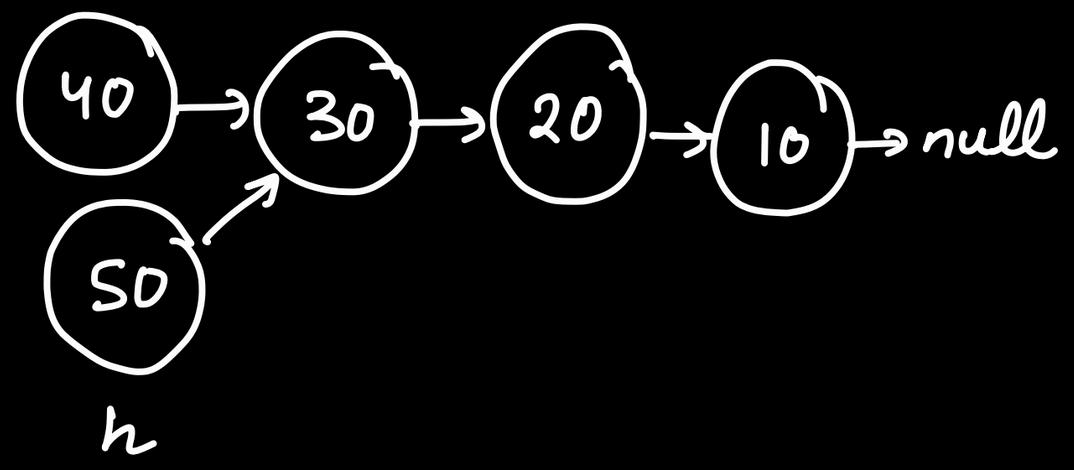
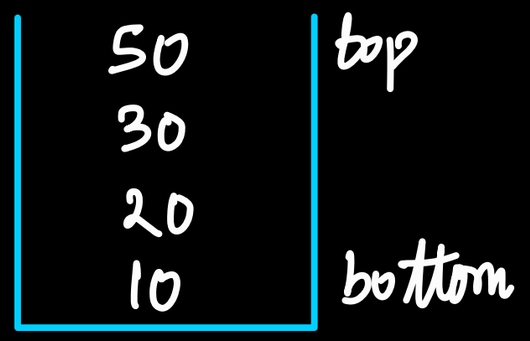
Q. Reverse the stack recursively.



Implementation of Stack via Linked-List

peek() → head.val

pop() → deleteAtHead

push() → addAtHead

size() → size